Machine Learning - Graduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Neural networks:
   (a) Optimize a convex objective function
   (b) Can only be trained with stochastic gradient descent
   (c) Can use a mix of different activation functions
   (d) None of the above

2. _ refers to a model that can neither model the training data nor generalize to new data.
   (a) good fitting
   (b) overfitting
   (c) underfitting
   (d) all of the above

3. Which of the following sentence is FALSE regarding regression?
   (a) It relates inputs to outputs.
   (b) It is used for prediction.
   (c) It may be used for interpretation.
   (d) It discovers causal relationships

4. What would you do in PCA to get the same projection as SVD?
   (a) Transform data to zero mean
   (b) Transform data to zero median
   (c) Not possible
   (d) None of these

5. A and B are two events. If P(A, B) decreases while P(A) increases, which of the following is true?
   (a) P(A|B) decreases
   (b) P(B|A) decreases
   (c) P(B) decreases
   (d) All of above

True / False (10 marks)

Answer True or False

6. True or False: ImageNet contains images of various resolutions, providing a diverse dataset for training deep learning models.
7. True or False: Random crop and horizontal flip are commonly used image data augmentation techniques for enhancing natural images.
8. True or False: A smaller number of test examples is sufficient to obtain statistically significant results when the error rate is smaller.
9. True or False: DenseNets generally require more memory than ResNets due to their high connectivity patterns.
10. True or False: Lasso regression is more appropriate for feature selection because it can shrink some coefficients to zero, effectively performing variable selection.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the joint probability of the variables H, U, P, and W within the specified Bayesian Network structure. Provide a detailed analysis of how to express this as a product of conditional probabilities.
12. Discuss the existential risks associated with artificial intelligence as articulated by Stuart Russell, analyzing their implications for future AI development and safety measures.

End of Paper